

UNDISTILLABLE: MAKING A NASTY TEACHER THAT CANNOT TEACH STUDENTS

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Motivation & Background

Knowledge Distillation (KD):

Technique to transfer knowledge from a large, complex teacher to a lightweight student. Commonly used to improve student model performance using soft output probabilities.

Unwanted consequence:

KD can inadvertently allow adversaries to clone models even if only black-box access is provided (via APIs/executable software).

Objective:

How to allow a teacher to perform normally while preventing its knowledge from being distilled.

The Nasty Teacher Concept

Train a “nasty teacher” that maintains high classification accuracy but delivers misleading soft outputs when used in distillation.

Defensive Strategy

Degrade student performance by “poisoning” the distilled knowledge while preserving the teacher’s own performance.

Practical Impact

Protects the model’s IP against unauthorized cloning even when distributed as a black box.

Key Insight

Up to date when the paper was released, every related work on Knowledge Distillation utilized the same training approach: given the outputs of the teacher's model, try to simultaneously maximize CE of student's model outputs with targets and KL-divergence with outputs of teacher's model.

Formally

Given the teacher model $f_{\theta_T}(\cdot)$ and the student model $f_{\theta_S}(\cdot)$, we aim at optimizing the following:

$$\min_{\theta_S} \sum_{x_i, y_i} \alpha \tau^2 \text{KL}(\sigma_\tau(f_{\theta_T}(x_i)) \| \sigma_\tau(f_{\theta_S}(x_i))) + (1 - \alpha) \text{CE}(\sigma_\tau(f_{\theta_S}(x_i)), y_i)$$

Self-Undermining Knowledge Distillation

Instead of training a normal model, the authors propose to integrate an adversarial (counteracting) model $f_{\theta_A}(\cdot)$ during teacher training. The objective of the teacher model transforms into the following:

$$\min_{\theta_T} \sum_{x_i, y_i} \text{CE}(\sigma_\tau(f_{\theta_S}(x_i), y_i) - \omega \tau^2 \text{KL}(\sigma_\tau(f_{\theta_T}(x_i)) \| \sigma_\tau(f_{\theta_S}(x_i))))$$

Results

Table 1: Experimental results on CIFAR-10.

Teacher network	Teacher performance	Students performance after KD			
		CNN	ResNetC-20	ResNetC-32	ResNet-18
Student baseline	-	86.64	92.28	93.04	95.13
ResNet-18 (normal)	95.13	87.75 (+1.11)	92.49 (+0.21)	93.31 (+0.27)	95.39 (+0.26)
ResNet-18 (nasty)	94.56 (-0.57)	82.46 (-4.18)	88.01 (-4.27)	89.69 (-3.35)	93.41 (-1.72)

Table 2: Experimental results on CIFAR-100.

Teacher network	Teacher performance	Students performance after KD			
		Shufflenetv2	MobilenetV2	ResNet-18	Teacher Self
Student baseline	-	71.17	69.12	77.44	-
ResNet-18 (normal)	77.44	74.24 (+3.07)	73.11 (+3.99)	79.03 (+1.59)	79.03 (+1.59)
ResNet-18 (nasty)	77.42(-0.02)	64.49 (-6.68)	3.45 (-65.67)	74.81 (-2.63)	74.81 (-2.63)
ResNet-50 (normal)	78.12	74.00 (+2.83)	72.81 (+3.69)	79.65 (+2.21)	80.02 (+1.96)
ResNet-50 (nasty)	77.14 (-0.98)	63.16 (-8.01)	3.36 (-65.76)	71.94 (-5.50)	75.03 (-3.09)
ResNeXt-29 (normal)	81.85	74.50 (+3.33)	72.43 (+3.31)	80.84 (+3.40)	83.53 (+1.68)
ResNeXt-29 (nasty)	80.26(-1.59)	58.99 (-12.18)	1.55 (-67.57)	68.52 (-8.92)	75.08 (-6.77)

Table 3: Experimental results on Tiny-ImageNet

Teacher network	Teacher performance	Students performance after KD			
		Shufflenetv2	MobilenetV2	ResNet-18	Teacher Self
Student baseline	-	55.74	51.72	58.73	-
ResNet-18 (normal)	58.73	58.09 (+2.35)	55.99 (+4.27)	61.45 (+2.72)	61.45 (+2.72)
ResNet-18 (nasty)	57.77 (-0.96)	23.16 (-32.58)	1.82 (-49.90)	44.73 (-14.00)	44.73 (-14.00)
ResNet-50 (normal)	62.01	58.01 (+2.27)	54.18 (+2.46)	62.01 (+3.28)	63.91 (+1.90)
ResNet-50 (nasty)	60.06 (-1.95)	41.84 (-13.90)	1.41 (-50.31)	48.24 (-10.49)	51.27 (-10.74)
ResNeXt-29 (normal)	62.81	57.87 (+2.13)	54.34 (+2.62)	62.38 (+3.65)	64.22 (+1.41)
ResNeXt29 (nasty)	60.21 (-2.60)	42.73 (-13.01)	1.09 (-50.63)	54.53 (-4.20)	59.54 (-3.27)

Results

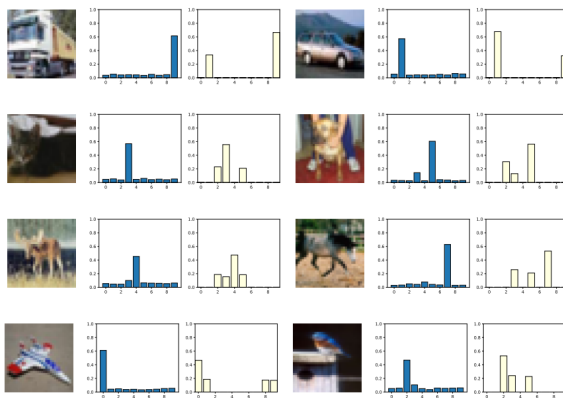
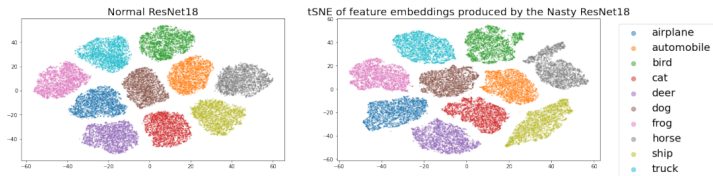
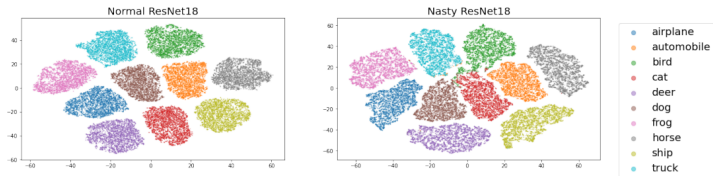


Figure 1: The visualization of logit responses after “temperature softmax” function. Each row represents two examples from CIFAR-10. The sampled images are shown in the 1st and the 4th columns. The 2nd and the 5th columns summarize the scaled output from the normal teacher. The 3rd and the 6th columns represent the scaled output from the nasty teacher

Results



(a) tSNE of feature embeddings before fully-connected layer. The dimension of feature embeddings is 512.



(b) tSNE of output logits

Figure 2: Visualization of tSNEs for both normal and nasty ResNet18 on CIFAR-10. Each dot represents one data point.

Summary

1. The paper introduced an interesting mechanism called "nasty teacher" to protect the models from illegal cloning.
2. The nasty teacher approach proved to be an effective defensive strategy against KD-based model cloning.
3. Empirical results show that the proposed method works on par compared to the original model.